Lecture 2: Pattern matching and ELIZA

PSYC 51.07: Models of language and communication

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Recap from Lecture 1

Key ideas

- **Consciousness is complex:** multiple types, hard to define
- **Language $\neq$ Thought:** but they interact in interesting ways
- **Grounding matters:** meaning comes from experience

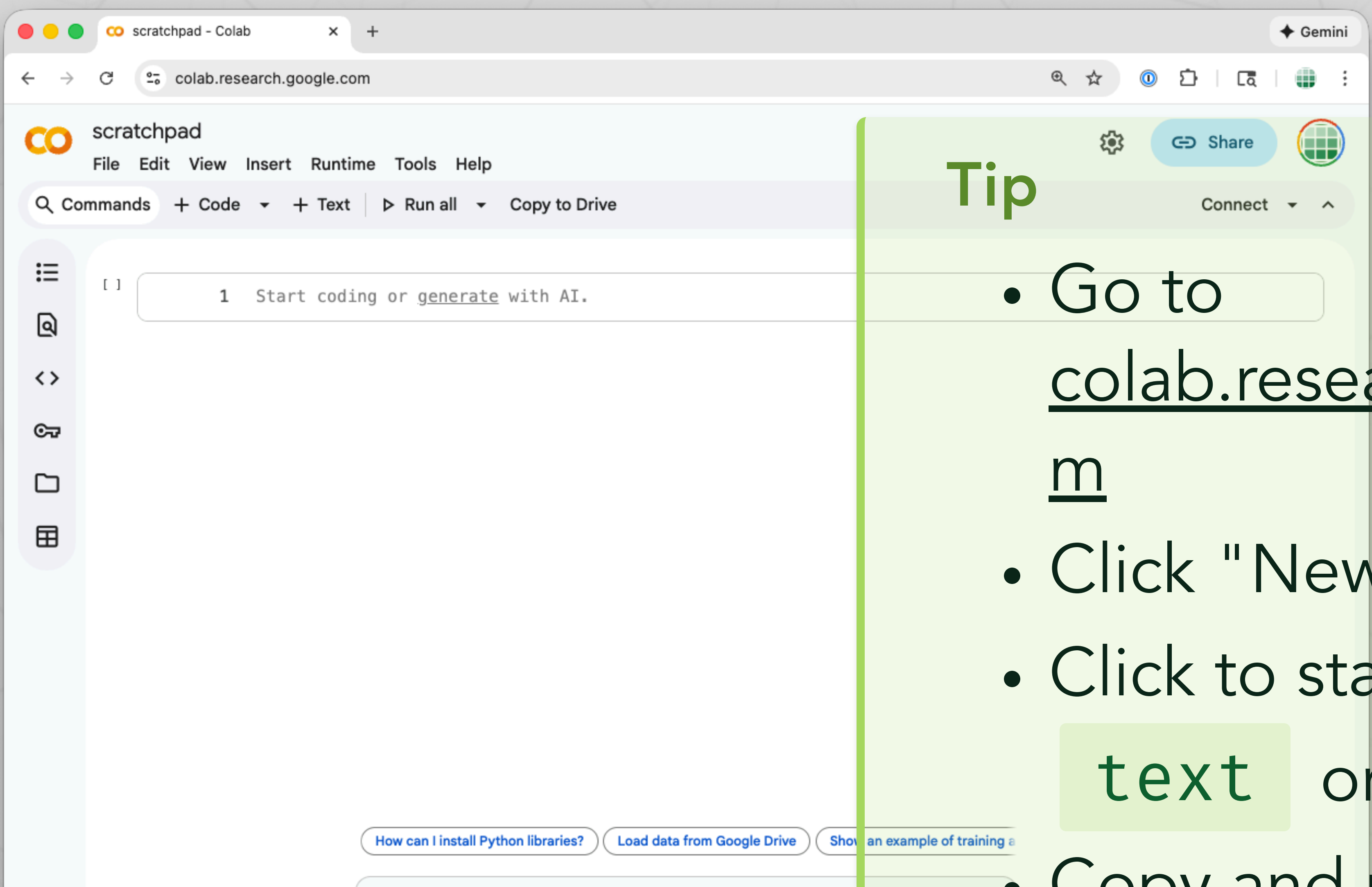
However...

Does the "experience" need to be first-hand? Can we "share" experience with another person? Or with a machine? What might that look like?

Today's focus

Pattern matching can create powerful *illusions* of understanding, even without any "real" comprehension.

Side note: follow along with [Google Colab](https://colab.research.google.com)!



Tip

- Go to colab.research.google.com
- Click "New notebook"
- Click to start new `text` or `code` cells
- Copy and paste code

Creating the illusion of experience and understanding

Humans

- Derive meaning from experience
- Connect words to memories, emotions, senses
- Understand context and nuance

Computers

- Manipulate strings (sequences of characters)
- Have no direct experience of the world
- Process symbols without inherent meaning

Text processing and string manipulation

- **Finding:** Locate patterns within text
- **Replacing:** Substitute one pattern for another
- **Extracting:** Pull out specific parts of text
- **Transforming:** Convert text to different formats

```
text = "Hello, h

# Finding
"how" in text

# Replacing
text.replace("yo

# Extracting
text.split(", ")
```

Text processing is the foundation of computational linguistics

User input → Pattern matching → Response generation → Output

Basic conversational AI pipeline

Note

Every conversational AI system, from ELIZA to ChatGPT, fundamentally processes text through some form of pattern matching, though with vastly different levels of sophistication.

Regular expressions

Regular expression (regex)

A sequence of characters that defines a search pattern. Regular expressions provide flexible, powerful pattern matching for text processing.

Key syntax

Symbol	Meaning	Example
`.`	Any character	`h.t` matches "hat", "hit", "hot"
`*`	Zero or more	`ab*c` matches "ac", "abc", "abbc"
`()`	Capture group	`(hello)` captures "hello"
`|`	Alternation	`cat|dog` matches "cat" or "dog"

Regular expressions in Python

```python
import re

text = "I am feeling very happy today"

# Simple pattern matching
if re.search(r"happy|sad|angry", text):
    print("Found an emotion!")

# Capture groups - extract parts of a match
match = re.search(r"I am feeling (.*?) today", text)
if match:
    emotion = match.group(1)  # "very happy"

# Substitution
new_text = re.sub(r"I am", "You are", text)
# "You are feeling very happy today"
```

Meet ELIZA: a computerized Rogerian therapist

Historical context

ELIZA was created by Joseph Weizenbaum at MIT in 1966. It was the first-ever interactive chatbot, and one of the first programs to attempt natural language processing. ELIZA plays the role of a Rogerian therapist, using simple pattern matching to simulate a conversation with a human patient.

Why a Rogerian therapist?

- Non-directive therapy style
- Reflects statements back to patient
- Asks open-ended questions

Key insight

- Rogerian style requires no real knowledge
- Simply reflect and rephrase
- Let the human do the

Chat with ELIZA

Try it!

Have a conversation with ELIZA. Try putting yourself into the "frame of mind" of someone from the 1960s who had never experienced a chatbot before, and likely who had only had limited (if any) exposure to computers. Take on the role of a "patient" seeking help from ELIZA in its role as a therapist. Then use your own (modern) knowledge and experiences to see where ELIZA breaks down.

- What does ELIZA do surprisingly well?
- What reveals its limitations?
- Can you "trick" ELIZA? How?
- What kinds of inputs break the illusion?
- **How do you think ELIZA *works*?**

The ELIZA effect

The ELIZA effect

The tendency to unconsciously assume that computer behaviors are analogous to human behaviors; to attribute human-like understanding to programs that merely simulate it.

Weizenbaum's observation

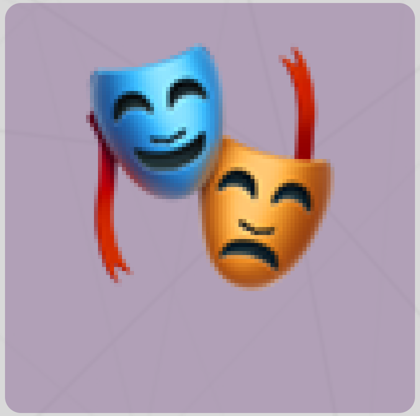
Weizenbaum was surprised (and disturbed) by how quickly users became emotionally involved with ELIZA. His secretary reportedly asked him to leave the room so she could have a private conversation with the program!

Why do we anthropomorphize machines?


Language cues


Pattern recognition


Social instincts


Theory of mind

> **Note**
> Humans are social creatures. We evolved to detect minds and intentions, and we often over-apply this tendency, even to clearly non-conscious entities.

Discussion

Reflect on your own experiences

Have *you* experienced the ELIZA effect with modern AI systems (ChatGPT, Claude, Siri, Alexa)?

- When have you felt like an AI "understood" you?
- Where do you think that illusion came from?
- What broke the illusion?
- What is the difference between *seeming* intelligent and *being* intelligent?
- How would we *know* if an AI truly understood us?

Reading: Weizenbaum (1966)

Required reading

Weizenbaum, J. (1966). ELIZA—A computer program for the study of natural language communication between man and machine. *Communications of the ACM*, 9(1), 36–45.

Pay special attention...

- How does ELIZA select responses?
- What are "scripts" in ELIZA's architecture?
- Why did Weizenbaum choose the DOCTOR script?
- What did Weizenbaum observe about user reactions?

Up next

Lecture 3 (Thursday X-hour)

How ELIZA *actually* works

- Complete architecture walkthrough
- Pattern matching and response selection
- The role of scripts and keywords
- We will build it ourselves!

Prepare for next time

- Finish reading Weizenbaum (1966)
- Play with the ELIZA demo
- Think about: what would *you* add to ELIZA?
- Read the Assignment 1 instructions

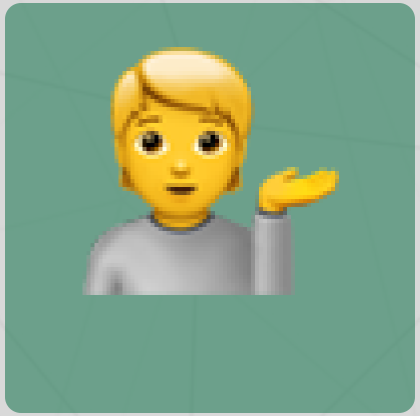
Key takeaways

1. **String manipulation is foundational:** all text-based AI builds on finding, replacing, and extracting patterns
2. **Regular expressions are powerful:** flexible pattern matching enables sophisticated text processing
3. **ELIZA demonstrated the power of simplicity:** a few clever rules can create convincing illusions
4. **The ELIZA effect is real:** we naturally anthropomorphize systems that use language
5. **Seeming $\neq$ Being:** appearing intelligent does not require actual understanding

Questions? Want to chat more?

Email me

Join our Discord



Come to office hours

Tip

Feeling lost? Want to make sure we cover something you're excited about? **Please** reach out if you have questions, comments, concerns, or just want to chat!